

TDT4195 Exercise 3

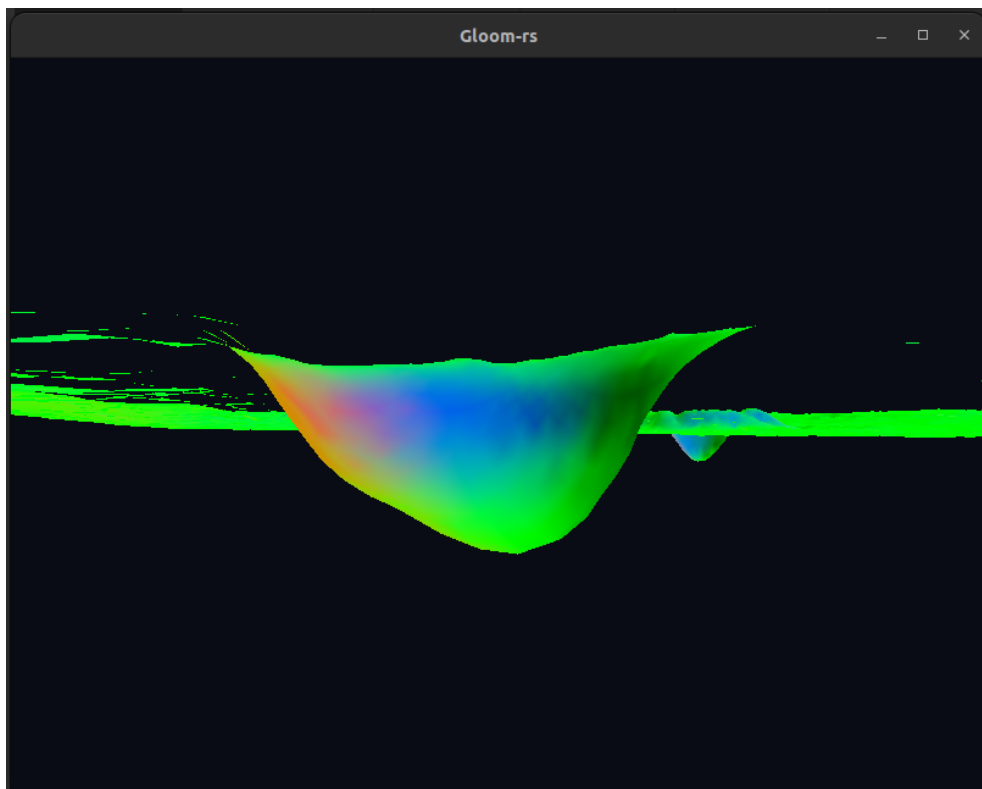
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1 Task 1: More polygons than you can shake a stick at

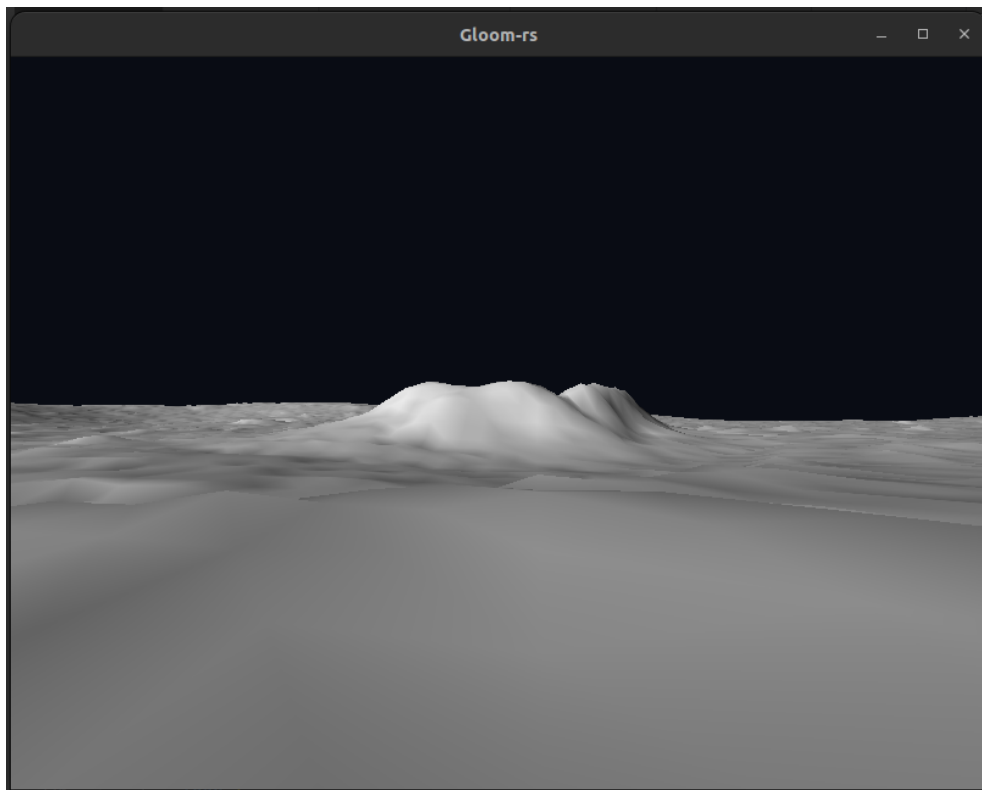
1.1 (c)



The camera coordinates to achieve this:

```
let mut cameraX = 250.0;  
let mut cameraY = 0.0;  
let mut cameraZ = -500.0;  
let mut angleX = 0.0;  
let mut angleY = 0.0;
```

1.2 (d)



2 Task 2: Helicopter Parenting

2.1 (c)



draw_scene function to achieve this:

```
unsafe fn draw_scene(node: &scene_graph::SceneNode,
    view_projection_matrix: &glm::Mat4,
    transformation_so_far: &glm::Mat4,
    shader: &shader::Shader)
{
    // logic before drawing the node

    // check if node drawable, set uniforms, bind vao, draw vao
    if node.index_count != -1 {
        let loc = shader.get_uniform_location("camera_transformation_matrix");
        gl::UniformMatrix4fv(loc, 1, gl::FALSE, view_projection_matrix.as_ptr());

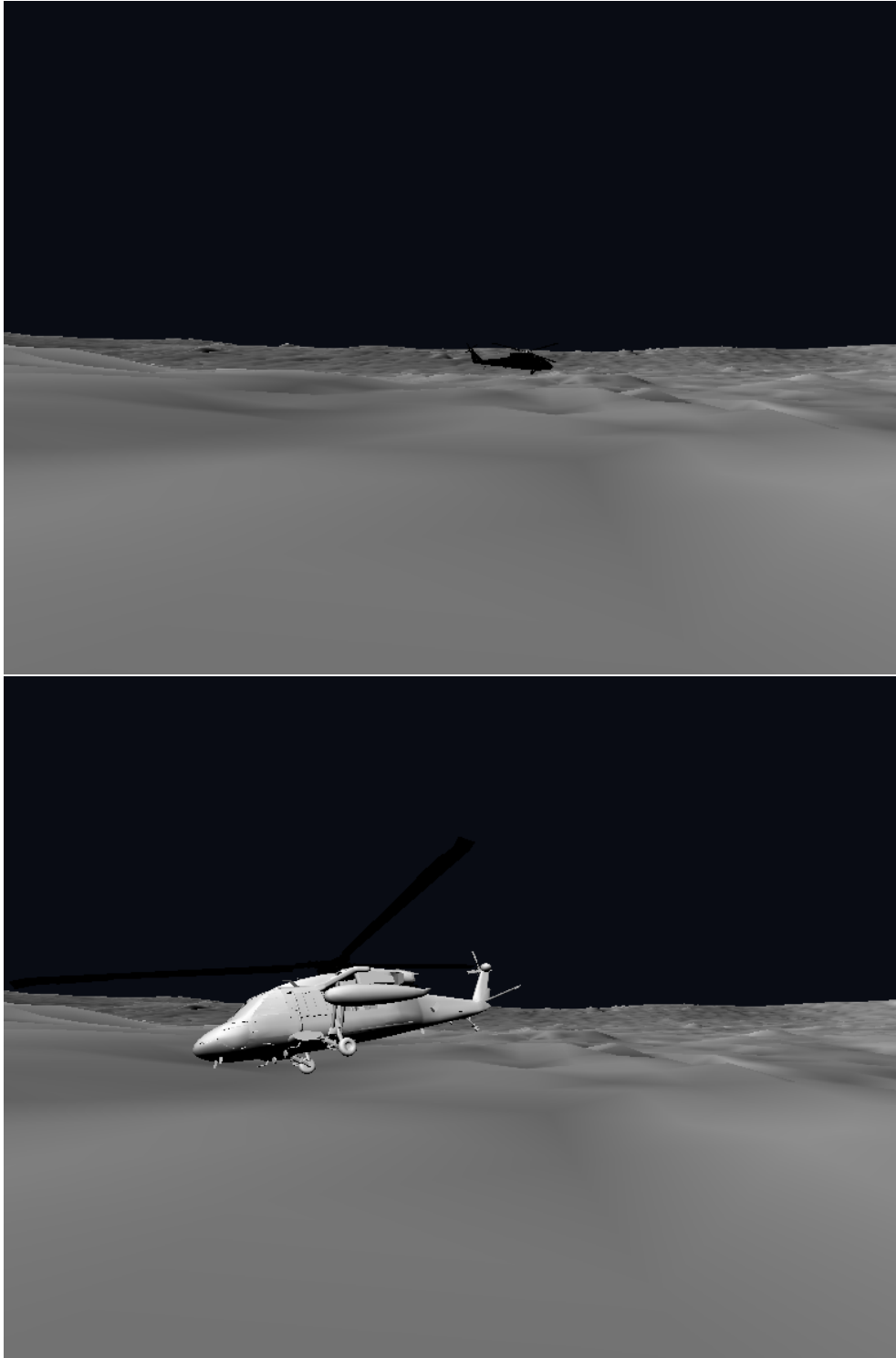
        gl::BindVertexArray(node.vao_id);

        gl::DrawElements(
            gl::TRIANGLES,
            node.index_count as i32,
            gl::UNSIGNED_INT,
            ptr::null(),
        );
    }
}
```

```
// Recurse
for child in &node.children {
    draw_scene(&child, view_projection_matrix, transformation_so_far, &shader);
}
}
```

3 Task 5: Help! My lighting is wrong!

3.1 (a)



As expected, the lighting doesn't change as the helicopter is moving.

3.2 (c)

Rotating the normals has been achieved by passing the model matrix as a uniform variable to the vertex shader:

```
let mut model_transform = transformation_so_far*local_transformation_matrix;

if node.index_count != -1 {
    let loc_transformation = shader.get_uniform_location("camera_transformation_matrix");
    gl::UniformMatrix4fv(loc_transformation, 1, gl::FALSE, model_view_projection.as_ptr());

    let loc_model = shader.get_uniform_location("model_matrix");
    gl::UniformMatrix4fv(loc_model, 1, gl::FALSE, model_transform.as_ptr());

    gl::BindVertexArray(node.vao_id);

    gl::DrawElements(
        gl::TRIANGLES,
        node.index_count as i32,
        gl::UNSIGNED_INT,
        ptr::null(),
    );
}
```

And modifying the vertex shader as shown below:

```
#version 430 core

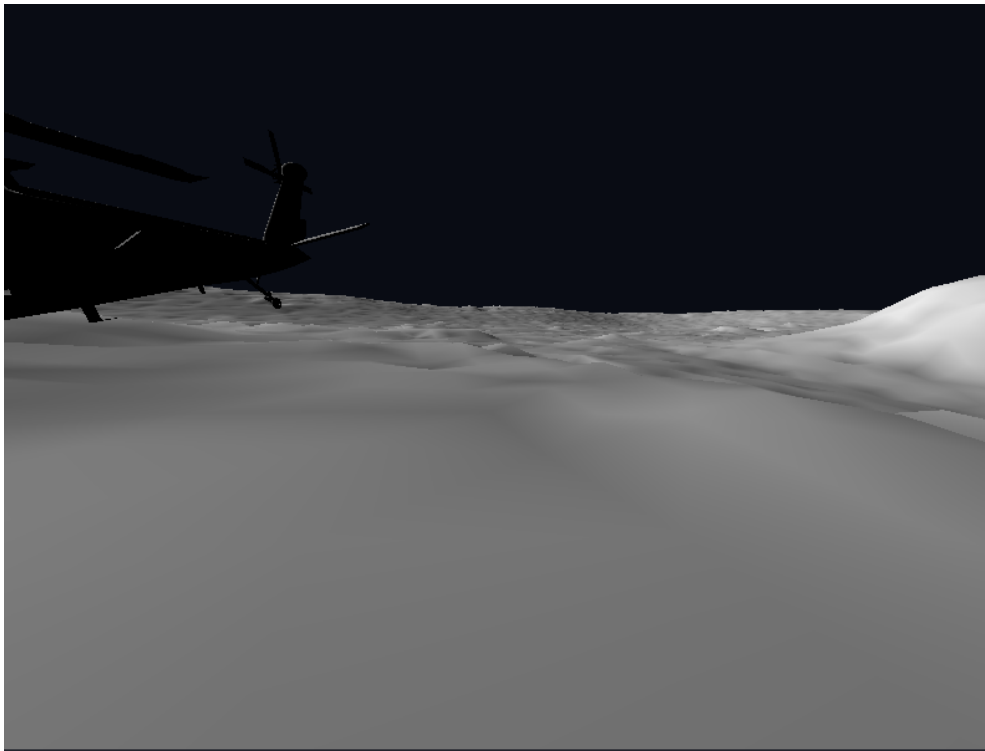
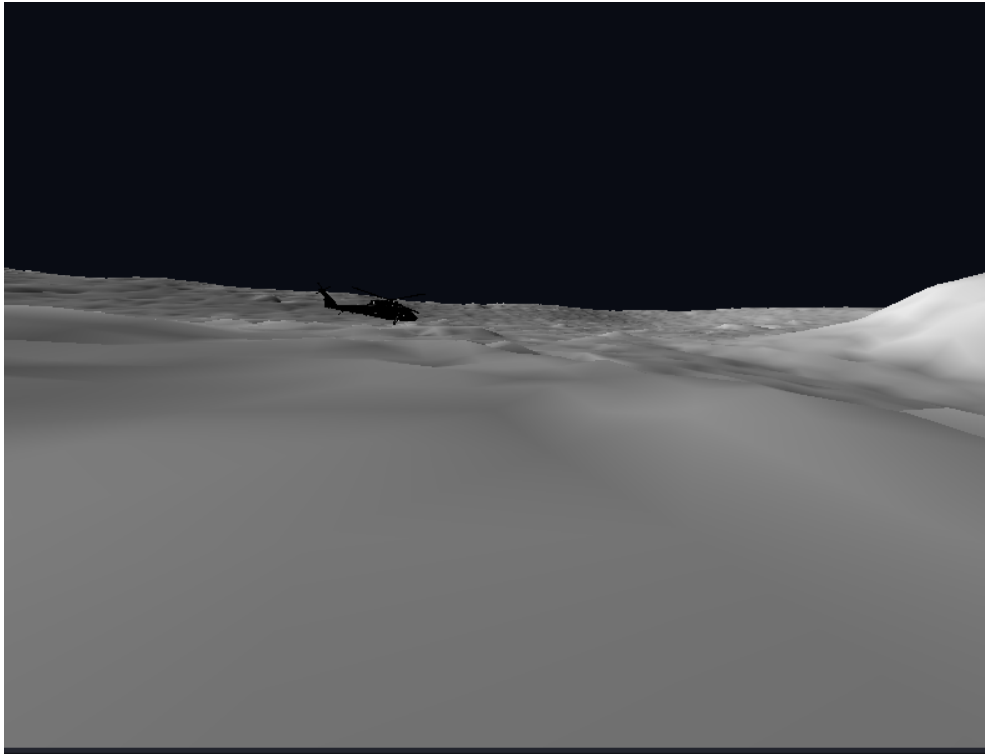
layout (location = 0) in vec3 position;
layout (location = 1) in vec3 normals;

uniform mat4x4 camera_transformation_matrix;
uniform mat4x4 model_matrix;

out VS_OUTPUT {
    vec3 color;
} OUT;

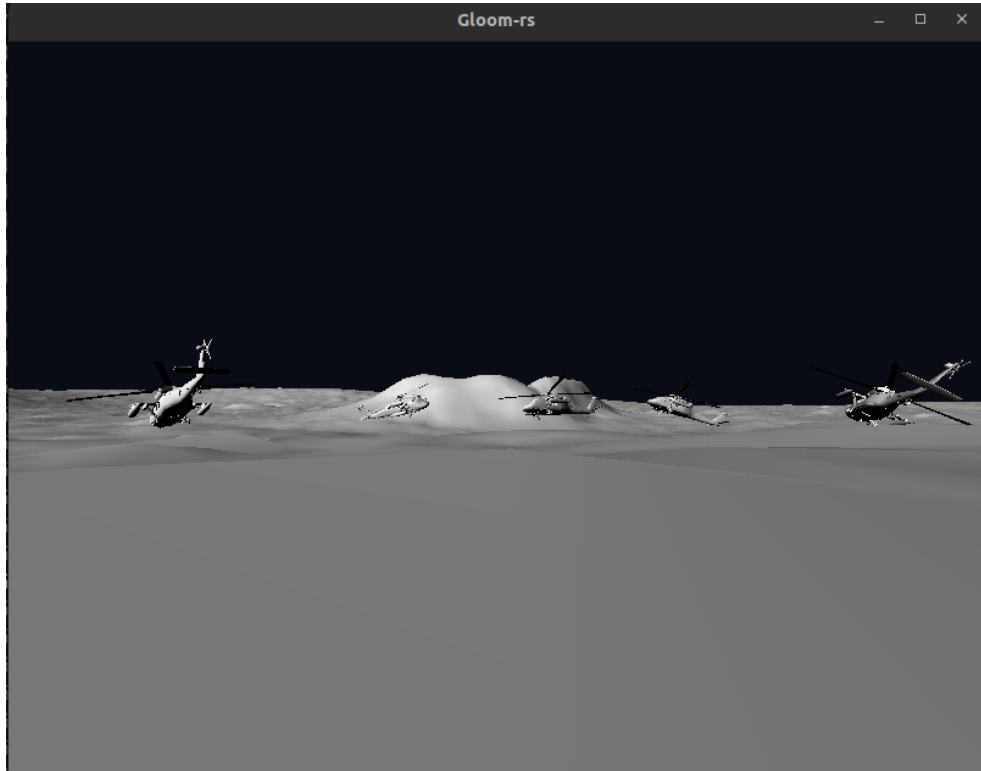
void main()
{
    vec4 temPos = vec4(position, 1.0f);
    vec4 newPosition = camera_transformation_matrix*temPos;
    gl_Position = newPosition;
    OUT.color = normalize(mat3(model_matrix)*normals);
}
```

This results in:



4 Task 6: Time to turn this thing up to 11 5

4.1 (a)



Loop to create the helicopters at the beginning of main:

```
for i in 0..5 {
    let mut terrain_node = scene_graph::SceneNode::from_vao(vao_terrain,
terrain_mesh.indices.len() as i32);
    let mut body_node = scene_graph::SceneNode::from_vao(vao_body,
helicopter_mesh.body.indices.len() as i32);

    let mut door_node = scene_graph::SceneNode::from_vao(vao_door,
helicopter_mesh.door.indices.len() as i32);

    let mut main_rotor_node = scene_graph::SceneNode::from_vao(vao_main_rotor,
helicopter_mesh.main_rotor.indices.len() as i32);

    let mut tail_rotor_node = scene_graph::SceneNode::from_vao(vao_tail_rotor,
helicopter_mesh.tail_rotor.indices.len() as i32);

    let mut scene_graph = scene_graph::SceneNode::new();

    body_node.add_child(&door_node);
    body_node.add_child(&main_rotor_node);
    body_node.add_child(&tail_rotor_node);
    terrain_node.add_child(&body_node);
}
```

```

scene_graph.add_child(&terrain_node);

// Setting the initial positions and rotations of every helicopter node
body_node.position = glm::vec3(0.0, 0.0, 0.0);
door_node.position = glm::vec3(0.0, 0.0, 0.0);
main_rotor_node.position = glm::vec3(0.0, 0.0, 0.0);
tail_rotor_node.position = glm::vec3(0.0, 0.0, 0.0);

body_node.rotation = glm::vec3(0.0, 0.0, 0.0);
door_node.rotation = glm::vec3(0.0, 0.0, 0.0);
main_rotor_node.rotation = glm::vec3(0.0, 0.0, 0.0);
tail_rotor_node.rotation = glm::vec3(0.0, 0.0, 0.0);

// Setting the reference points for every helicopter node
body_node.reference_point = glm::vec3(0.0, 0.0, 0.0);
door_node.reference_point = glm::vec3(1.2, 0.0, -1.0);
main_rotor_node.reference_point = glm::vec3(0.0, 2.0, 0.0);
tail_rotor_node.reference_point = glm::vec3(0.35, 2.3, 10.4); // this is correct

helicopters.push(scene_graph);
}

Loop to draw the helicopters:
for i in 0..5 {
    helicopters[i].get_child(0).get_child(0).get_child(1).rotation =
        glm::vec3(0.0, 10.0*elapsed, 0.0);
    helicopters[i].get_child(0).get_child(0).get_child(2).rotation =
        glm::vec3(10.0*elapsed, 0.0, 0.0);

    let heading = toolbox::simple_heading_animation(elapsed + ((i*10) as f32));
    helicopters[i].get_child(0).get_child(0).position.x = heading.x;
    helicopters[i].get_child(0).get_child(0).position.z = heading.z;
    helicopters[i].get_child(0).get_child(0).rotation.x = heading.roll;
    helicopters[i].get_child(0).get_child(0).rotation.y = heading.yaw;
    helicopters[i].get_child(0).get_child(0).rotation.z = heading.pitch;

    draw_scene(&helicopters[i], &camera_transformation_matrix, &glm::identity(),
        &simple_shader);
}

```

5 Optional Task: Finding Easter Egg

